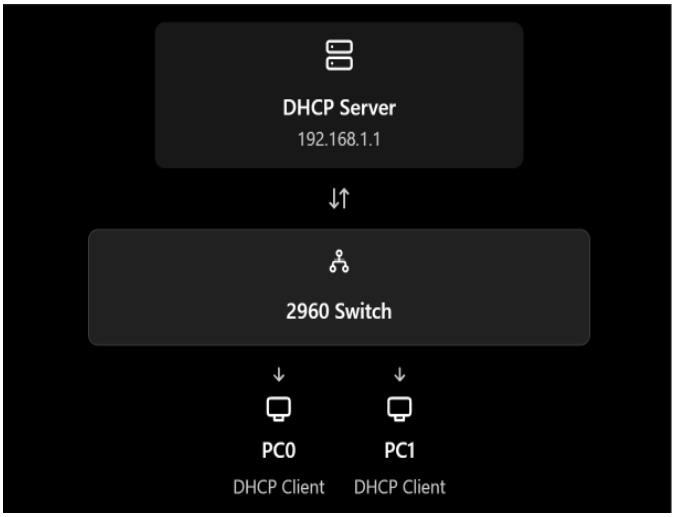
Day 2: DHCP ConfigurationTheory (30 min)

1. In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Aim: To configure a DHCP server in Cisco Packet Tracer and verify that two client PCs automatically receive different IP addresses from the DHCP pool.

1. Network topology



2. Procedure

Step 1: Create the topology

1. Open Cisco Packet Tracer.
2. Add one Server, one 2960 Switch, and two PCs (PC0 and PC1).
3. Connect all three end devices to the switch using Copper Straight-Through cables.
4. Wait until the link indicators turn green.

Step 2: Configure the DHCP server

1. Click the Server and open Desktop → IP Configuration.
2. Select Static and enter:

Setting	Value
IP Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.254

3. Open Services → DHCP.
4. Turn DHCP On and configure the pool:

Setting	Value
Pool Name	LAN_POOL
Default Gateway	192.168.1.254
DNS Server	8.8.8.8
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Number of Users	50

5. Click Add or Save, depending on your Packet Tracer version.

Step 3: Configure PC0 and PC1

For each PC:

1. Click the PC.
2. Open Desktop → IP Configuration.
3. Select DHCP instead of Static.
4. Wait for the DHCP request to complete.
5. Check that each PC receives an IP address automatically.

Step 4: Verify the IP addresses

On PC0 and PC1, open Desktop → Command Prompt and execute:

ipconfig

Example expected output:

Device	IPv4 Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0

Note: These are expected example addresses. Your actual results may differ depending on the order in which the PCs request addresses.

To test connectivity, run this command on PC0:

ping 192.168.1.11

If PC1 received 192.168.1.11, successful replies indicate that communication between the PCs is working.

3. Observation

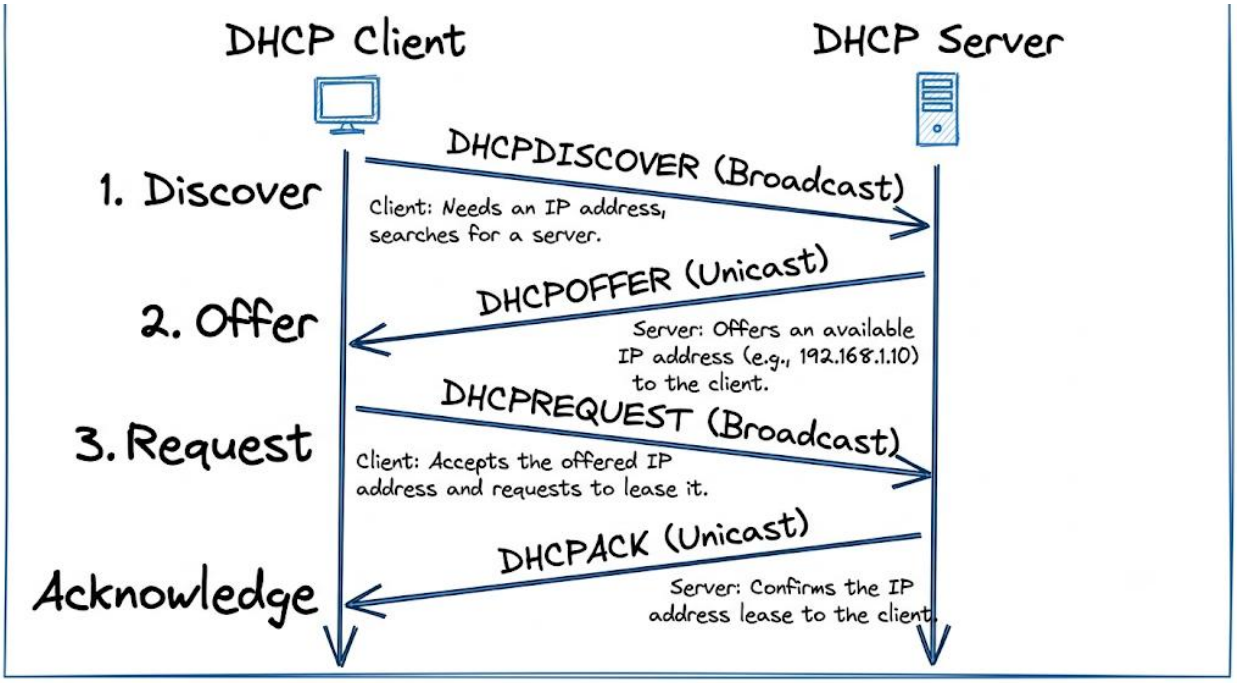
- The DHCP server was configured with a pool of IP addresses.
- PC0 and PC1 obtained their IP addresses automatically.
- Both PCs received different IP addresses within the configured DHCP pool.
- The ipconfig command was used to verify the assigned IP addresses.

4. Result

The DHCP server was successfully configured in Cisco Packet Tracer. Both client PCs received different IP addresses automatically from the DHCP pool, demonstrating the working of Dynamic Host Configuration Protocol (DHCP).

3. **Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.

Hint: Use arrows to show the direction of each message between client and server.**



3.Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Procedure

1. Open your Cisco Packet Tracer file from Experiment 1.
2. Click the DHCP Server → Services → DHCP.
3. Change the DHCP pool settings to allow only one IP address:
    - Start IP Address: 192.168.1.10
    - Subnet Mask: 255.255.255.0
    - Maximum Number of Users: 1
4. Click Save or Add, as applicable, to apply the settings.
5. Click PC0 and PC1, open Desktop → IP Configuration, and select DHCP on both PCs.
6. Observe the assigned IP addresses.

Expected observation

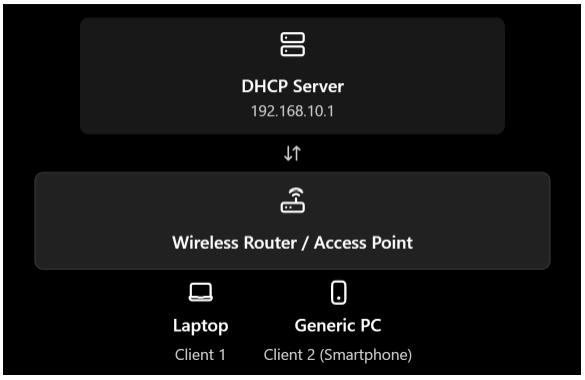
Device	Expected result
PC0	Receives 192.168.1.10
PC1	Fails to obtain an address from the DHCP pool

The first PC to receive the available lease gets the IP address; the other PC may show a DHCP request failure or an automatic/private address, depending on the simulation.

4.Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Aim: To simulate a Zomato-style restaurant network in Cisco Packet Tracer where a laptop and a smartphone receive IP addresses automatically from a DHCP server.

1. Network topology



Scenario: A customer visits a restaurant, connects their laptop and smartphone to the restaurant Wi-Fi, and opens Zomato to browse menus or place an order.

2. Procedure

Step 1: Add devices

Open Cisco Packet Tracer and add:

- One Server (DHCP server)
- One wireless router or access point
- One Laptop
- One generic PC representing a smartphone

Connect the server to the router using an Ethernet cable. Connect the laptop and generic PC to the wireless network if your chosen devices support wireless connectivity. Otherwise, use wired connections to demonstrate DHCP.

Step 2: Configure the DHCP server

Click Server → Desktop → IP Configuration and set:

Setting	Value
IP Address	192.168.10.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.254

Next, open Services → DHCP, turn DHCP On, and configure the pool:

Setting	Value
Pool Name	RESTAURANT_WIFI
Start IP Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.254
DNS Server	8.8.8.8
Maximum Users	50

Click Add or Save.

Step 3: Connect the clients

1. Configure the wireless router with the SSID Zomato_Restaurant_WiFi.
2. Set a Wi-Fi password if wireless connectivity is supported.
3. Connect the laptop and the generic PC to the same network.
4. On each client, open Desktop → IP Configuration and select DHCP.

Step 4: Verify the assigned IP addresses

Open Desktop → Command Prompt on each client and run:

ipconfig

Expected results:

Device	Assigned IP address	Subnet mask
Laptop	192.168.10.10	255.255.255.0
Generic PC (Smartphone)	192.168.10.11	255.255.255.0

These are example addresses; your actual assignments may differ depending on which client requests an address first.

3. Why is DHCP useful in a restaurant?

DHCP automatically assigns IP addresses to customers' devices when they connect to restaurant Wi-Fi. This avoids manually configuring every laptop or smartphone, prevents duplicate IP assignments within the same network, and makes it easier for many customers to connect and use services such as Zomato.